

GENERAL

1. ORIGINAL SITE SURVEY PREPARED BY: PRO SURVEY
CONTOUR INTERVAL: Xxm
DATUM:
VERTICAL PMXXX, RLX.XXX AHD
HORIZONTAL XXXX
REAL PROPERTY DESCRIPTION:
LOT XXX ON SLXXXX
PARISH OF XXXX
COUNTY OF XXXX
2. THESE DRAWINGS ARE TO BE READ IN CONJUNCTION WITH THE PROJECT GEOTECHNICAL REPORT BY XXX GEOTECHNICAL REPORT No.XXX DATED XX 20XX.
3. CONTRACTOR MUST VERIFY ALL DIMENSIONS AND EXISTING LEVELS ON SITE PRIOR TO COMMENCEMENT OF WORK.
4. EXISTING SERVICES HAVE BEEN PLOTTED FROM SUPPLIED DATA AND AS SUCH THEIR ACCURACY CANNOT BE GUARANTEED. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO ESTABLISH THE LOCATION AND LEVEL OF ALL EXISTING SERVICES PRIOR TO COMMENCEMENT OF ANY WORK. ANY DISCREPANCIES SHALL BE REPORTED TO THE SUPERINTENDENT.
5. THE CONTRACTOR SHALL ARRANGE ALL SURVEY SETOUT BY A LICENSED LAND SURVEYOR. CONFIRM SETOUT WITH ARCHITECT.
6. ALL SERVICE TRENCHES UNDER VEHICULAR PAVEMENTS SHALL BE BACKFILLED WITH APPROVED FILL MATERIAL AND COMPACTED TO 100% STANDARD MAXIMUM DRY DENSITY IN ACCORDANCE WITH AS 1289 E1.1.
7. ON COMPLETION OF SERVICES INSTALLATION, ALL DISTURBED AREAS SHALL BE RESTORED TO ORIGINAL, INCLUDING KERBS, FOOTPATHS, CONCRETE AREAS, GRAVEL AREAS, GRASSED AREAS AND ROAD PAVEMENTS.
8. WHERE NEW WORK ABUTS EXISTING, THE CONTRACTOR SHALL ENSURE THAT A SMOOTH EVEN PROFILE FREE FROM ABRUPT CHANGES, IS OBTAINED.
9. CARE TO BE TAKEN WHEN EXCAVATING ENERGEX, TELSTRA AND ORIGIN GAS SERVICES. NO MECHANICAL EXCAVATION TO BE UNDERTAKEN OVER ENERGEX, TELSTRA AND ORIGIN GAS; HAND EXCAVATE IN THESE AREAS. LIAISE WITH RELEVANT AUTHORITY.
10. ALL WORKS SHALL BE IN ACCORDANCE WITH RELEVANT COUNCIL STANDARDS, DETAILS SHOWN ON THE DRAWINGS, THE SPECIFICATION AND THE DIRECTIONS OF THE ARCHITECT.
11. THE CONTRACTOR SHALL NOT DISTURB ANY EXISTING SURVEY BENCH MARKS OR REFERENCE MARKS (UNLESS INDICATED FOR REMOVAL) WITHOUT DEPT. OF NATURAL RESOURCES APPROVAL IN WRITING.
12. THE CONTRACTOR SHALL OBTAIN ALL RELEVANT AUTHORITIES APPROVALS. (PRIOR TO COMMENCEMENT AND UPON COMPLETION).
13. WORK ON EXISTING EXTERNAL STORMWATER SERVICES BY CONTRACTOR ARE SUBJECT TO INSPECTION AND APPROVAL BY COUNCIL.
14. THE CONTRACTOR SHALL LOCATE AND CONFIRM LEVELS OF ALL EXISTING SERVICE CONNECTIONS PRIOR TO COMMENCING WORK.
15. THE CONTRACTOR SHALL REFER TO LANDSCAPE DRAWINGS FOR DETAILS OF TREATMENT OF BATTERS, SWALE DRAINS AND ANY OTHER AREAS DISTURBED AS PART OF CIVIL WORKS.
16. ROAD OPENINGS:
THE CONTRACTOR'S ATTENTION IS DRAWN TO THE REQUIREMENTS OF COUNCIL WHERE THE WORK IS LOCATED IN A ROAD RESERVE OR CROSSES THE PAVEMENT OF A PUBLIC ROADWAY OR FOOTWAY. OBTAIN A CONSENT COVERING THE NECESSARY OPENINGS AND COMPLY WITH ALL THE CONDITIONS COVERING THE ISSUE OF SUCH A CONSENT. A CHARGE IS MADE FOR PERMITS ISSUED IN RESPECT OF PAVEMENT OPENINGS. CONTRACTOR TO OBTAIN DETAILS FROM COUNCIL, MEET ALL REQUIREMENTS AND PAYMENTS, AND CARRY OUT ALL LIAISON AT NO EXTRA COST. ALL PAVEMENT SURFACING IS TO BE UNDERTAKEN BY THE CONTRACTOR. ALL PAVEMENT REINSTATEMENT TO BE UNDERTAKEN BY THE CONTRACTOR.
17. REFER ALSO TO HYDRAULIC, LANDSCAPING, MECHANICAL AND ELECTRICAL CONSULTANTS' DRAWINGS FOR ADDITIONAL SITE SERVICES TO BE CO-ORDINATED FOR CONSTRUCTION UNDER PROPOSED CARPARK AND ROAD PAVEMENTS.
18. A GEOTECHNICAL INVESTIGATION FOR THE ORIGINAL SITE HAS BEEN PREPARED, REFER TO THE PROJECT GEOTECHNICAL REPORT. THE CONTRACTOR SHALL ENSURE THAT THEY ARE FAMILIAR WITH THE CONTENTS OF THIS REPORT AND THE REPORT DOES FORM PART OF THE CONTRACT.
19. ALL CONSTRUCTION WORK SHALL BE CARRIED OUT SO THAT AT ANY TIME ADJOINING PROPERTY OWNERS ARE NOT DEPRIVED OF AN ALL-WEATHER ACCESS OR SUBJECTED TO ADDITIONAL STORM WATER RUN-OFF DURING THE PERIOD OF CONSTRUCTION.
20. CONTRACTOR SHALL BE RESPONSIBLE FOR REPAIR OF ANY DAMAGE TO COUNCIL'S INFRASTRUCTURE. SUCH REPAIR OR REINSTATEMENT TO BE CARRIED OUT IMMEDIATELY TO THE SATISFACTION OF THE SUBDIVISION AND DEVELOPMENT MANAGER OF COUNCIL.
21. SITE INSPECTIONS REQUIRED AS SPECIFIED.
22. FOR CIVIL SETOUT INFORMATION CONTACT WSP (PH 3854 6200) AND AN AUTOCAD DRAWING FILE WILL BE ISSUED FOR SITE SETOUT SUBJECT TO CONDITIONS.
23. CONTRACTOR IS TO ALLOW TO INCLUDE ANY REQUIRED COUNCIL INSPECTION AND COMPLIANCE FEES IN THE CONTRACT.

EARTHWORKS

1. FOLLOWING DEMOLITION OF EXISTING STRUCTURES, ALL VEGETATION, SOILS CONTAINING ORGANIC AND DELETERIOUS MATERIAL, CONCRETE SLABS, CONCRETE BUILDING FOOTINGS AND PAVEMENT AREAS SHALL BE STRIPPED AND REMOVED FROM SITE.
2. ALL EXISTING TREE STUMPS AND ROOTS SHALL BE GRUBBED OUT TO A MINIMUM DEPTH OF 750mm.
3. DEPRESSIONS FORMED BY REMOVAL OF UNDERGROUND ELEMENTS AND OR WEAKENED AREAS SHOULD BE CLEANED OUT AND FILLED WITH COMPACTED IMPORTED FILL MATERIAL.
4. EXPOSED CONSTRUCTION AREAS SHALL BE PROOF ROLLED USING A VEHICLE WITH A LOADING OF AT LEAST 5 TONNES PER AXLE IN THE PRESENCE OF THE NOMINATED GEOTECHNICAL TESTING AUTHORITY AND A REPRESENTATIVE OF WSP. AREAS DEMONSTRATING EXCESSIVE MOVEMENT SHALL BE TREATED IF POSSIBLE, OR REMOVED TO COMPETENT MATERIAL AND REPLACED WITH COMPACTED IMPORTED FILL MATERIAL. CONTRACTOR TO ALLOW A PROVISIONAL SUM FOR REMOVAL OF UNSUITABLE MATERIAL AND REPLACEMENT WITH COMPACTED IMPORTED FILL MATERIAL. ALLOW A PROVISIONAL QUANTITY OF XXXm³
5. ALL SOFT, WET OR UNSUITABLE MATERIAL IS TO BE REMOVED AND REPLACED WITH APPROVED MATERIAL SATISFYING THE REQUIREMENTS LISTED BELOW FOR IMPORTED FILL MATERIAL.
6. PLACE IMPORTED FILL MATERIALS SATISFYING THE REQUIREMENTS LISTED BELOW TO NOMINATED BULK EARTHWORKS LEVELS ON BULK EARTHWORKS PLAN.
7. ALL IMPORTED FILL MATERIAL SHALL BE FROM A SOURCE APPROVED BY THE SUPERINTENDENT AND SHALL COMPLY WITH THE FOLLOWING:
 - a) FREE FROM ORGANIC AND PERISHABLE MATTER
 - b) MAXIMUM PARTICLE SIZE 100mm
 - c) PLASTICITY INDEX MAXIMUM 15%
 - d) LIQUID LIMIT, MAX OF 45%
 - e) PASSING 19mm SIEVE, MAX 80%
 - f) MINIMUM CBR 10%
 - g) SHRINK/SWELL INDEX, MAX OF 1.0%
8. CONTRACTOR TO PROVIDE CERTIFICATION TO THE SUPERINTENDENT FROM NATA REGISTERED LAB THAT THE FILL MATERIAL IS FREE OF ACTUAL OR POTENTIAL ACID SULFATE SOILS AND MEETS CURRENT DPI REGULATIONS FOR THE CONTROL OF RED IMPORTED FIRE ANTS (RIFA).
9. HAND ENGINEERED FILL MATERIAL SHALL BE PLACED IN MAXIMUM 200mm THICK LAYERS AND COMPACTED AT OPTIMUM MOISTURE CONTENT (+ OR -2%) TO ACHIEVE THE FOLLOWING MINIMUM COMPACTION VALUES:

LOCATION	COHESIVE SOILS	COHESIONLESS SOILS
BUILDING AREAS		
• FOUNDATION SUPPORT	98%	80%
• FLOOR SLABS	98%	80%
PAVEMENT AREAS		
• TOP 300mm BELOW SUBGRADE LEVEL	98%	80%
• BELOW THE TOP 300mm	95%	75%
LANDSCAPE AREAS	95%	75%

WHERE THE ABOVE VALUES ARE DETERMINED IN ACCORDANCE WITH AS 1289 5.1.1 (STANDARD COMPACTION) FOR COHESIVE SOILS AND AS 1289 5.6.1 (STANDARD METHOD) FOR COHESIONLESS SOILS.
10. ALL EARTHWORKS OPERATIONS SHALL BE CARRIED OUT WITH 'LEVEL 1' ENGINEERING SUPERVISION IN ACCORDANCE WITH APPENDIX B OF AS 3798. THE GEOTECHNICAL TESTING AUTHORITY SHALL BE FROM THE NOMINATED PANEL OF PROVIDERS WITHIN THE CIVIL SPECIFICATION. THE NOMINATED GEOTECHNICAL TESTING AUTHORITY SHALL PROVIDE CERTIFICATION AS NOTED IN THE GEOTECHNICAL REPORT THAT ALL GENERAL EARTHWORKS OPERATIONS HAVE BEEN CARRIED OUT IN ACCORDANCE WITH THE DRAWINGS AND SPECIFICATIONS AND THE CONTROLLED FILL IS SUITABLE FOR PURPOSE WITH A MINIMUM 150 kPa BEARING CAPACITY UNDER THE BUILDING. ALL CERTIFICATIONS RECOMMENDED IN THE PROJECT GEOTECHNICAL REPORT ARE TO BE PROVIDED BY THE CONTRACTOR.
11. THE CONTRACTOR SHALL PROGRAMME THE EARTHWORKS OPERATION SO THAT THE WORKING AREAS ARE ADEQUATELY DRAINED DURING THE PERIOD OF CONSTRUCTION. THE SURFACE SHALL BE GRADED AND SEALED OFF TO REMOVE DEPRESSIONS. ROLLER MARKS AND SIMILAR WHICH WOULD ALLOW WATER TO POND AND PENETRATE THE UNDERLYING MATERIAL. ANY DAMAGE RESULTING FROM THE CONTRACTOR NOT OBSERVING THESE REQUIREMENTS SHALL BE RECTIFIED BY THE CONTRACTOR AT THEIR COST. CONTRACTOR IS TO ALSO NOTE THAT THE EXISTING SOILS INCLUDING NATURAL SILTY SAND AND HIGH PLASTICITY CLAYS ARE LIKELY TO BE SUSCEPTIBLE TO SOFTENING WHEN WET. TO THE POINT WHERE THE SITE MAY BECOME UNTRAFFICABLE. THE CONTRACTOR IS TO PROVIDE TEMPORARY HAUL ROADS WHERE NECESSARY TO ALLOW ACCESS TO THE WORKS AREAS.
12. TESTING OF THE SUBGRADE SHALL BE CARRIED OUT IN ACCORDANCE WITH AS 3798 BY AN APPROVED NATA REGISTERED LABORATORY AT THE CONTRACTOR'S EXPENSE.
13. ALL BASE AND SUB-BASE MATERIALS SHALL BE PLACED IN MAXIMUM 150mm THICK LAYERS AND COMPACTED AT OPTIMUM MOISTURE CONTENT OF (+ OR -2%) TO ACHIEVE A DRY DENSITY IN ACCORDANCE WITH AS 1289 5.2.1 OF NOT LESS THAN THE FOLLOWING:

LAYER	MODIFIED DRY DENSITY
BASE COURSE	98% MODIFIED (AS 1289 5.2.1)
SUB-BASE COURSE	95% MODIFIED (AS 1289 5.2.1)
14. CONTRACTOR TO ENSURE VIA THE APPROPRIATE SELECTION OF FILL MATERIAL AND EARTHWORKS THAT, UPON COMPLETION OF THE EARTHWORKS, THAT THE BUILDING PLATFORM HAS A CLASS 'M' CLASSIFICATION IN ACCORDANCE WITH AS 2870.
15. ANY SURPLUS EXCAVATED MATERIAL SHALL BE DISPOSED OF OFF SITE AT AN APPROVED TIP SITE. CONTRACTOR TO ALLOW TO COMPLY WITH CURRENT DPI REGULATIONS FOR THE CONTROL OF RED IMPORTED FIRE ANTS.
16. ANY CONTAMINATED MATERIAL SHALL BE MANAGED AND DISPOSED OF ACCORDING TO THE CURRENT LOCAL COUNCIL AND EPA REQUIREMENTS INCLUDING CERTIFICATION.
17. THE USE OF POTABLE WATER IS NOT PERMITTED IN ACTIVITIES ASSOCIATED WITH ROAD AND PAVEMENT CONSTRUCTION. THE COMPACTION OF FILL MATERIAL OR DUST SUPPRESSION CONTRACTOR TO UTILISE RECYCLED OR SUITABLE NON POTABLE WATER FOR THESE ACTIVITIES WITHIN THE CONTRACT. USE OF RECYCLED WATER MUST BE IN ACCORDANCE WITH THE REQUIREMENTS OF COUNCIL.

EXISTING SERVICES AND FEATURES

1. THE CONTRACTOR SHALL ALLOW FOR THE CAPPING OFF, EXCAVATION AND REMOVAL IF REQUIRED OF ALL EXISTING SERVICES IN AREAS AFFECTED BY THE WORKS WITHIN THE CONTRACT AREA, AS SHOWN ON THE DRAWINGS UNLESS DIRECTED OTHERWISE BY THE SUPERINTENDENT. ALL TO REGULATORY AUTHORITY STANDARDS AND APPROVAL.
2. THE CONTRACTOR SHALL ENSURE THAT AT ALL TIMES SERVICES TO ALL BUILDINGS NOT AFFECTED BY THE WORKS ARE NOT DISRUPTED.
3. PRIOR TO COMMENCEMENT OF ANY WORKS THE CONTRACTOR SHALL OBTAIN THE SUPERINTENDENT'S APPROVAL FOR THEIR PROGRAMME FOR THE RELOCATION/CONSTRUCTION OF TEMPORARY SERVICES.
4. EXISTING BUILDINGS, EXTERNAL STRUCTURES, AND TREES SHOWN ON THESE DRAWINGS ARE FEATURES EXISTING PRIOR TO ANY SEPARATE AND PRECEDING DEMOLITION WORKS.
5. CONTRACTOR SHALL CONSTRUCT TEMPORARY SERVICES TO MAINTAIN EXISTING SUPPLY TO BUILDINGS REMAINING IN OPERATION DURING WORKS TO THE SATISFACTION AND APPROVAL OF THE SUPERINTENDENT. ONCE DIVERSION IS COMPLETE AND COMMISSIONED THE CONTRACTOR SHALL REMOVE ALL SUCH TEMPORARY SERVICES AND MAKE GOOD TO THE SATISFACTION OF THE SUPERINTENDENT.
6. INTERRUPTION TO SUPPLY OF EXISTING SERVICES SHALL BE DONE SO AS NOT TO CAUSE ANY INCONVENIENCE TO THE ADJACENT RESIDENCES. CONTRACTOR TO GAIN APPROVAL OF THE SUPERINTENDENT FOR TIME OF INTERRUPTION.

STORMWATER

1. ALL 300 DIA. DRAINAGE PIPES AND LARGER SHALL BE CLASS '3" APPROVED SPIGOT AND SOCKET RCP PIPES WITH RUBBER RING JOINTS, (U.N.O.). ALL DRAINAGE PIPES UP TO AND INCLUDING 225 DIA. SHALL BE SEWER GRADE uPVC WITH SOLVENT WELD JOINTS, (U.N.O.).
2. EQUIVALENT STRENGTH FRC PIPES MAY BE USED.
3. ALL PIPE JUNCTIONS UP TO AND INCLUDING 450 DIA. AND ALL TAPERS SHALL BE VIA PURPOSE MADE FITTINGS.
4. MINIMUM GRADE TO STORMWATER LINES TO BE 1%. (U.N.O.).
5. CONTRACTOR TO SUPPLY AND INSTALL ALL FITTINGS AND SPECIALS INCLUDING VARIOUS PIPE ADAPTORS TO ENSURE PROPER CONNECTION TO DISSIMILAR PIPEWORK.
6. ALL CONNECTIONS TO EXISTING DRAINAGE PITS SHALL BE MADE IN A TRADESMAN-LIKE MANNER AND THE INTERNAL WALL OF THE PIT AT THE POINT OF ENTRY SHALL BE CEMENT RENDERED TO ENSURE A SMOOTH FINISH. THE CONTRACTOR IS TO REPLACE DOWNSTREAM RECEIVING STORMWATER STRUCTURES, SHOULD THE STRUCTURE BE DAMAGED DURING INSTALLATION.
7. WHERE SUBSOIL DRAINAGE LINES PASS UNDER FLOOR SLABS AND VEHICULAR PAVEMENTS UNSLOTTED uPVC SEWER GRADE PIPE SHALL BE USED.
8. PROVIDE 3.0m LENGTH OF 100 DIA. SUBSOIL DRAINAGE PIPE WRAPPED IN FABRIC SOCK, AT UPSTREAM END OF EACH STORMWATER STRUCTURE (UNO)
9. BEDDING SHALL BE TYPE H1 (U.N.O.) IN ACCORDANCE WITH CURRENT AUSTRALIAN STANDARDS.
10. AT CONNECTION TO PITS, INVERT OF SUBSOIL DRAINS TO BE ABOVE MAIN STORMWATER PIPE OVERT.
11. EQUIVALENT STRENGTH PRECAST STORMWATER PITS AND MANHOLES MAY BE USED.
12. PROVIDE CCTV INSPECTION AND REPORT OF ALL INSTALLED STORMWATER PIPES AT COMPLETION OF WORKS.
13. ALL PIPS TO BE INSTALLED IN ACCORDANCE WITH THE LOCAL AUTHORITY STANDARDS AND METHODS.

GROUTED ROCK PITCHING

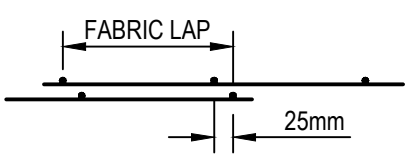
1. ROCK SHALL BE OF SIZE (MINIMUM DIMENSION) NOT LESS THAN 100mm AND NOT GREATER THAN 400mm UNO.
2. ROCK SHALL BE PLACED IN TWO LAYERS OF NOMINAL TOTAL THICKNESS 300mm.
3. WHERE ROCK PITCHING IS CONSTRUCTED FOR BATTER PROTECTION, THE TOE ROCKS SHALL BE MINIMUM SIZE 200mm.
4. THE FIRST LAYER OF ROCK SHALL BE PLACED ON SAND BEDDING OF MINIMUM 50mm THICKNESS.
5. ROCKS SHALL BE PLACED SO AS TO FORM IRREGULAR JOINTS. ALL ROCKS SHALL BE INTERLOCKED AND WEDGED WITH SIMILAR SIZE ROCK AS NECESSARY SO THAT NO SINGLE ROCK CAN BE EASILY DISLODGED AND NO LARGE VOIDS REMAIN BETWEEN PLACED ROCK.
6. THE VOIDS BETWEEN ROCKS AT THE EXPOSED SURFACE SHALL BE FILLED WITH CEMENT MORTAR.
7. CEMENT MORTAR FOR BEDDING SHALL CONSIST OF ONE PART (BY VOLUME) OF GENERAL PORTLAND CEMENT TO THREE PARTS CLEAN FINE SAND. THE MORTAR SHALL BE ABLE TO RETAIN ITS SHAPE AND NOT FLOW LIKE A LIQUID.
8. ROCK SHALL BE OF SOUND COMPOSITION AND NOT DISINTEGRATE IN WATER.
9. PLACED ROCK TO SIT PROUD OF MORTAR BEDDING NOMINAL 75mm TO PROVIDE ROUGH SURFACE PROFILE FOR ENERGY DISSIPATION.

REINFORCED CONCRETE

- GENERAL
1. CARRY OUT ALL CONCRETE WORK IN ACCORDANCE WITH AS 3600 AND THE SPECIFICATION. KEEP A COPY OF THE DOCUMENTS ON SITE.
 2. VERIFY ALL SETTING OUT DIMENSIONS WITH THE SUPERINTENDANT AND/OR THE SURVEYOR.
 3. DO NOT OBTAIN DIMENSIONS BY SCALING THE DRAWINGS.
- CONCRETE
4. ALL WORKMANSHIP AND MATERIALS SHALL COMPLY WITH AS 3600 CURRENT EDITIONS WITH AMENDMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
 5. CONCRETE CHARACTERISTICS:

ELEMENT	COMPRESSIVE STRENGTH F _c - 28 DAYS (MPa)	SLUMP (mm)	MAX AGG SIZE (mm)
ALL KERBS, PITS ETC	32	80	20
FOOTPATH	32	80	20
CARPARK PAVEMENT	32	65	20
OSD TANK BASE AND LID	40	80	20
 6. NO 'BRECCIA' TYPE AGGREGATE IS TO BE USED.
 7. CEMENT TO BE TYPE SL TO AS 3972 U.N.O. MAX CONCRETE SHRINKAGE TO BE 600 MICRON TO AS1012.
 8. NO ADMIXTURES ARE TO BE USED WITHOUT THE APPROVAL OF THE ENGINEERS.
 9. A VIBRATOR IS TO BE USED FOR THE COMPACTION OF ALL CONCRETE.
 10. ALL CONCRETE PLACEMENT TO UTILISE ALIPHATIC ALCOHOL APPLIED TO THE CONCRETE AFTER THE INITIAL SCREED. CONCRETE SHALL BE CURED IN ACCORDANCE WITH AS 3600 FOR A MINIMUM OF 7 DAYS AFTER POURING, UTILISING AN APPROVED, NON-OIL BASED CURING COMPOUND OR THE COVERING OF THE ENTIRE SLAB WITH A CLEAR PLASTIC SHEET.
 11. CONSTRUCTION JOINTS WHERE NOT SHOWN SHALL BE LOCATED TO THE APPROVAL OF THE ENGINEERS.
 12. ALL CONCRETE SHALL BE SUBJECT TO PROJECT CONTROL SAMPLE AND TESTING TO AS 3600.
 13. THE CONTRACTOR MAY ADD AN APPROVED SUPERPLASTICISING ADMIXTURE TO INCREASE SLUMP OF PAVEMENT CONCRETE TO NOT MORE THAN 100mm. QUALITY AND MIXING PROCEDURE TO BE IN ACCORDANCE WITH THE MANUFACTURERS WRITTEN INSTRUCTIONS.
- FINISHING
14. THE CONCRETE SHALL BE SCREEDED TO THE REQUIRED CROSS SECTION PROFILE, FREE OF DEPRESSIONS AND HIGH AREAS TO SATISFY THE REQUIREMENTS OF AN INITIAL FINISH.
 15. FINAL FINISHING SHALL COMMENCE ONLY AS SOON AS THE WATER SHEEN HAS LEFT THE PAVEMENT SURFACE AND NOT IN ANY AREA WHERE THERE IS FREE SURFACE WATER. REFER ARCHITECTURAL SPECIFICATIONS FOR FINISHES TO SURFACES.
- REINFORCEMENT
16. COVER TO REINFORCEMENT FOR CORROSION PROTECTION SHALL BE AS DETAILED OR AS A MINIMUM 40mm TO TOP REINFORCEMENT, 50mm TO BOTTOM REINFORCEMENT.
 17. PIPES OR CONDUITS SHALL NOT BE PLACED WITHIN THE COVER TO REINFORCEMENT.
 18. REINFORCEMENT QUALITY
 19. 19.2 BAR REINFORCEMENT

SYMBOL	BAR SHAPE	STRENGTH GRADE (MPa)	DUCTILITY CLASS	TO COMPLY WITH AUST. STANDARD
N	DEFORMED RIBBED BAR	500	NORMAL	AS 4671
R	PLAIN ROUND BAR	250	NORMAL	AS 4671
- ALL BARS TO BE "N" UNLESS NOTED OTHERWISE THE NUMBER FOLLOWING THE SYMBOL IS THE NOMINAL BAR SIZE IN mm. FOR EXAMPLE, N16 DENOTES A DEFORMED RIBBED BAR, OF GRADE 500MPa NORMAL DUCTILE STEEL, WITH A NOMINAL 16mm DIAMETER.
*NOTE: Y BARS MAY BE REPLACED WITH N BARS OF SAME SIZE.
- | SYMBOL | DENOTES | STRENGTH
GRADE (MPa) | DUCTILITY
CLASS | TO COMPLY
WITH AUST.
STANDARD |
|--------|--|-------------------------|--------------------|-------------------------------------|
| RL | RECTANGULAR MESH OF DEFORMED RIBBED BARS | 500 | LOW | AS 4671 |
| SL | SQUARE MESH OF DEFORMED RIBBED BARS | 500 | LOW | AS 4671 |
- THE NUMBERS FOLLOWING THE SYMBOL DENOTE THE PRODUCT CODE. FOR EXAMPLE, SL92 DENOTES A SQUARE MESH OF 9mm (NOMINAL DIAMETER) DEFORMED RIBBED BARS AT 200mm CENTRES, OF GRADE 500MPa LOW DUCTILITY STEEL.
19. REINFORCEMENT IS REPRESENTED DIAGRAMMATICALLY; IT IS NOT NECESSARILY SHOWN IN TRUE PROJECTION.
 20. SPLICES IN REINFORCEMENT SHALL BE MADE ONLY IN THE POSITION SHOWN. THE WRITTEN APPROVAL OF THE ENGINEER SHALL BE OBTAINED FOR ANY OTHER SPLICE
 21. WELDING OF REINFORCEMENT WILL BE PERMITTED ONLY AFTER APPROVAL BY THE ENGINEER.
 22. SITE BENDING OF N BARS SHALL BE DONE COLD WITH POWER OR MECHANICAL BENDING TOOLS.
 23. FABRIC LAPS


 24. FABRIC TO EXTEND 70 WITH A CROSSWIRE ONTO WALLS UNLESS NOTED OTHERWISE.
 25. ALL REINFORCEMENT SHALL BE SECURELY SUPPORTED IN CORRECT POSITION DURING CONCRETING BY APPROVED PLASTIC BAR CHAIRS UNO. MAXIMUM CENTERS OF SUPPORTING CHAIRS SHALL BE 600mm FOR FABRIC, 600mm FOR BARS UP TO 12mm DIAMETER AND 900mm FOR BARS 16mm AND GREATER.

ASPHALTIC CONCRETE

- GENERAL
1. ASPHALTIC CONCRETE MIX DESIGN, MANUFACTURE, PLACING AND COMPACTION SHALL BE IN ACCORDANCE WITH THE SPECIFICATION AND AS2150 - 2005 ASPHALT - A GUIDE TO GOOD PRACTICE.
- PAVEMENT PREPARATION
2. THE EXISTING SURFACE TO BE SEALED SHALL BE DRY AND BROOMED BEFORE COMMENCEMENT OF WORK TO ENSURE COMPLETE REMOVAL OF ALL SUPERFICIAL FOREIGN MATTER.
 3. ALL DEPRESSIONS OR UNEVEN AREAS ARE TO BE TACK-COATED AND BROUGHT UP TO GENERAL LEVEL OF PAVEMENT WITH ASPHALTIC CONCRETE BEFORE LAYING OF MAIN COURSE.
- TACK COAT
4. THE WHOLE OF THE AREA TO BE SHEETED WITH ASPHALTIC CONCRETE SHALL BE LIGHTLY AND EVENLY COATED WITH RAPID SETTING BITUMEN.
- JOINTS
5. THE NUMBER OF JOINTS BOTH LONGITUDINAL AND TRANSVERSE SHALL BE KEPT TO A MINIMUM.
 6. THE DENSITY AND SURFACE FINISH AT JOINTS SHALL BE SIMILAR TO THOSE OF THE REMAINDER OF THE LAYER.
- COMPACTION
7. ALL COMPACTION SHALL BE UNDERTAKEN USING SELF PROPELLED ROLLERS.
 8. INITIAL ROLLING SHALL BE COMPLETED BEFORE THE MIX TEMPERATURE FALLS BELOW 105°C.
 9. SECONDARY ROLLING SHALL BE COMPLETED BEFORE THE MIX TEMPERATURE FALLS BELOW 60°C.
- FINISHED PAVEMENT PROPERTIES
10. FINISHED SURFACES SHALL BE SMOOTH, DENSE AND TRUE TO SHAPE AND SHALL NOT VARY MORE THAN THE SPECIFIED TOLERANCES.

CONCRETE PAVEMENT JOINT

1. WHERE NOMINATED, JOINTS TO BE SAWN AS SOON AS CONCRETE HAS HARDENED SUFFICIENTLY THAT IT WILL NOT BE DAMAGED BY SAWING. IF AN UNPLANNED CRACK OCCURS THE CONTRACTOR SHALL REPLACE WHOLE SLABS EITHER SIDE OF THE UNPLANNED CRACK, UNLESS DIRECTED OTHERWISE.
2.
 - a) PROVIDE JOINTS AS DETAILED
 - b) ALL LONGITUDINAL JOINTS SHALL BE FORMED AND INCLUDE TIE BARS OR DIAMOND DOWELS AS SPECIFIED. ALL TRANSVERSE CONSTRUCTION JOINTS SHALL BE FORMED AND INCLUDE PLATES OR DIAMOND DOWELS AS SPECIFIED.
 - c) UNO BOND BREAKER TO BE PROPRIETARY DANLEY PLATE AND DIAMOND DOWEL COVERS.
3. DOWELS, PLATES AND TIE BARS SHALL BE:
 - STRAIGHT
 - TO LENGTH SPECIFIED
 - CLEAN AND FREE FROM MILL SCALE, RUST AND OIL
 - SAWN TO LENGTH NOT CROPPED
 - HOT DIPPED GALVANISED AFTER SAWING TO LENGTH
4. DIMENSIONS OF SEALANT RESERVOIR DEPENDANT ON THE SEALANT TYPE ADOPTED. ENGINEERS APPROVAL TO BE OBTAINED FOR SEALANT AND RESERVOIR DIMENSIONS AND DETAIL PROPOSED BY THE CONTRACTOR. REFER TO DETAILS FOR TYPICAL ARRANGEMENT AND SEALANT.
5. PRIOR TO THE PLACEMENT OF CONCRETE IN THE ADJACENT SLAB, SELF 10mm ABELFLEX WITH RIPOFF STRIP SHALL BE ADHERED TO THE ALREADY CAST AND CLEANED CONCRETE FACE USING AN APPROVED WATERPROOF ADHESIVE. ADHESIVE SHALL BE LIBERALLY APPLIED TO THE FULL FACE OF THE CONCRETE SLAB TO BE COVERED BY THE FILLER, AND ON THE FULL FACE OF THE FILLER TO BE ADHERED.
6. REFER TO EARTHWORKS NOTES FOR PREPARATION OF SUB-BASE AND SUB-GRADE.
7. CONTRACTION, LONGITUDINAL, ISOLATION AND CONSTRUCTION JOINTS SHALL BE LOCATED AS SPECIFIED.
8. SAWCUTTING SHALL COMMENCE AS EARLY AS POSSIBLE UTILISING SOFTCUT TO AVOID UNCONTROLLED SHRINKAGE CRACKING, BUT NOT SO EARLY AS TO CAUSE UNACCEPTABLE RAVELING OF JOINT EDGES. PROGRAM SITE AND TIME OF CONCRETE POURS TO PERMIT SAWING TO BE CARRIED OUT WITHOUT CAUSING NOISE POLLUTION.
9. CONCURRENTLY WITH THE SAWING OPERATIONS THE SAWN GROOVE SHALL BE WASHED FREE OF MATERIAL.
10. SAWN GROOVES FORMING THE RESERVOIR FOR A JOINT SEALANT SHALL BE FORMED IN TWO STAGES. THE SECOND SAWCUT SHALL BE MADE AS LATE AS POSSIBLE.

DANLEY DOWEL JOINT

1. WHERE NOMINATED THE CONTRACTOR IS TO ALLOW FOR ALL COMPONENTS OF DANLEY DOWEL JOINT SYSTEM INCLUDING TEMPLATES & PEGGING TO ENSURE THAT ALL DOWEL BARS REMAIN IN THE CORRECT ALIGNMENT & POSITION REFER DETAILS.

LINEMARKING

1. ALL PAVEMENT MARKING, RRPMS, GUIDEPOSTS AND TRAFFIC SIGNS TO BE INSTALLED IN ACCORDANCE WITH AS1742.1 AND THE COUNCIL STANDARDS, GUIDELINES AND SPECIFICATIONS.
2. ALL REDUNDANT LINEMARKINGS TO BE REMOVED BY ABRASIVE BLASTING (INCLUDING WATER BLASTING) OR BY COLD PLANING OR GRINDING.
3. PROVIDE REFLECTORISED PAINT TO MEDIAN ISLAND KERBS IN ACCORDANCE WITH COUNCIL STANDARDS AND SPECIFICATIONS.
4. ALL INTERNAL CARPARK LINE MARKING SHALL BE 90mm WIDE UNO AND BE COMPLETED USING WHITE REFLECTORISED PAINT UNO IN ACCORDANCE WITH AS 1742 AND AS SPECIFIED IN THE ARCHITECTURAL SPECIFICATION.
5. THE INTERNATIONAL SYMBOL OF ACCESS SHALL BE WHITE ON A BLUE BACKGROUND IN ACCORDANCE WITH AS 1428. THE BLUE SHALL BE B21, ULTRAMARINE OF AS 2700, OR SIMILAR APPROVED.

PI	30012008	RO	DRAFT 60% DESIGN DEVELOPMENT	JC	JC
REV	DATE	BY	DESCRIPTION	CHK	APP
DRAWING STATUS:			PRELIMINARY		



http://www.wsp.com

CLIENT:



ARCHITECT:



PROJECT:

CQU ROCKHAMPTON TAFE

TITLE:

GENERAL NOTES

SCALE @ A1:	CHECKED:	APPROVED:
NTS	JC	JC
PROJECT NUMBER:	DRAWN:	DATE:
PS229518	RO	1/30/2026
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